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A newly discovered PAX7 variant associated with craniofacial duplication: a case report

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Reporting Summary

Through whole-exome sequencing (WES), this investigation identifies a PAX7 variant of uncertain significance (VUS) in a patient with partial craniofacial duplication, providing preliminary molecular genetic evidence for this extremely rare anomaly and highlighting the need for functional validation. In summary, this case report lays the groundwork for future research into the genetic basis of craniofacial duplication, highlighting PAX7 as a candidate gene worthy of further investigation. By combining functional validation, large-cohort genetics, and clinical translation, we can advance our understanding of this rare disorder, improve prenatal detection, and optimize patient outcomes. Ultimately, these efforts may shed light on broader principles of craniofacial development and inform the management of other congenital anomalies.

Abstract

Background: Diprosopus (craniofacial duplication) is an exceedingly rare craniomaxillofacial

dysmorphism, considered a subgroup of conjoined twins, with phenotypes ranging from partial structural duplication to complete dicephalus. Its embryogenesis and molecular mechanisms remain poorly understood. This study describes a case of partial dentofacial duplication and reports novel genetic findings to explore potential etiology.

Case presentation: A newborn Kazak male was referred for a prenatal-detected maxillary mass. Physical examination revealed left complete unilateral cleft lip and a 2.5×3.0 cm soft tissue mass (accessory upper lip with a central sinus tract) originating from the ipsilateral upper lip. CT confirmed duplicated, translocated maxillary arches and widened nasal bone. The patient underwent two surgeries (at 11 months and 4 years of age) for cleft lip repair, duplicated maxilla excision, and bony contouring. Long-term follow-up showed unimpaired feeding and speech. Whole-exome sequencing (WES) of the patient and mother identified a shared heterozygous missense variant: PAX7 c.1033G>C (p.Ala345Pro, NM_001135254.1), classified as a variant of uncertain significance (VUS) per ACMG/AMP guidelines.

Conclusions: This is the first report of a PAX7 VUS associated with craniofacial duplication. PAX7 plays critical roles in craniofacial development, and the conserved Ala345 residue suggests potential functional impact. While causality is not established, this finding links PAX7 to a rare duplication phenotype, providing a preliminary molecular basis for future prenatal screening and mechanistic studies. Additional functional validation and larger cohort studies are needed to confirm pathogenicity.

Keywords: Birth defects, Craniofacial duplication, Diprosopus, PAX7, Variant of uncertain significance, Prenatal diagnosis, Clinical management

Introduction

Diprosopus, also known as craniofacial duplication, is an extremely rare congenital malformation, with fewer than 35 cases reported globally since 1900 [1]. Its incidence is estimated at <1/1,000,000 live births, with a female-to-male sex ratio of 1.4:1 [1, 2]. Phenotypes range from isolated duplication of facial elements (e.g., tongue, lip, maxilla) to complete dicephalus (two heads on one neck) [2, 3]. Common features include macrostomia, stomatodeal structure duplication, cleft lip/palate, and orbital hypertelorism [1–4]. Partial duplication most frequently involves the oral cavity, maxilla, or mandible [1, 5].

The etiology of craniofacial duplication remains controversial. Proposed mechanisms include duplication of cells at the anterior termination of the notochordal process (initiating

facial-oral element duplication) [3, 5], abnormal neural crest cell migration, or disruptions in early embryogenic signaling pathways (e.g., SHH, TWIST1) [6]. However, molecular genetic evidence for most cases is lacking, with no definitive disease-causing genes identified to date [2, 3]. Here, we report a case of partial craniofacial duplication (maxillary and upper lip involvement) in a Kazak male infant, with genetic analysis identifying a PAX7 variant of uncertain significance (VUS), and discuss its potential role in disease pathogenesis.

Case Presentation

For the description of the clinical case, we followed the CARE guidelines and the CARE checklist is provided as an additional file ([Supplemental Material 1](#)) [7].

A newborn Kazak male was referred to our institution following prenatal ultrasonography (third trimester) revealing a maxillary mass. The differential diagnosis at referral included teratoma, ectopic sinus, congenital cyst, foregut duplication, or fibrous dysplasia (rationale: prenatal radiological features of bony expansion and soft tissue mass overlapped with fibrous dysplasia). The patient had no family history of craniofacial deformities, and no in-utero teratogen exposure was reported.

At 11 months of age, physical examination revealed left complete unilateral cleft lip and a painless, soft, irregular 2.5×3.0 cm mass arising from the ipsilateral upper lip ([Appendix Fig. 1A](#)). A 15 mm-deep sinus tract was present at the mass center, pedicled to the upper gingiva with no oral cavity communication. The maxillary alveolar ridge was widened, and a cleft was noted at the nasal base with a horizontal left nostril ([Appendix Fig. 1B](#)). Ocular examination showed a shorter palpebral fissure and longer interocular axial length; other head and neck findings were unremarkable.

Head and neck CT demonstrated abnormally widened nasal bone, left complete cleft lip, irregular soft tissue from the left upper lip, and duplicated, midline-translocated maxillary arches ([Appendix Figs. 2–3](#)). No aggressive osseous features or periosteal reaction were seen. Given incomplete deciduous dentition development, primary cleft lip repair was performed via modified West China method [8] to avoid tooth germ injury.

At 4 years of age, the patient presented with mastication difficulties. Clinical examination showed nasolabial deformation from primary repair and duplicated maxillary dental arches with erupted deciduous teeth ([Appendix Fig. 4](#)). Preoperative CT confirmed two parallel upper dental arches with fully erupted deciduous teeth and duplicated permanent tooth germs ([Fig. 1A](#)).

Three-dimensional CT reconstruction was used to design customized surgical guides (Fig. 1B and Appendix Videos 1–2). The patient underwent duplicated maxilla excision, dental arch bony contouring, and soft tissue defect closure with adjacent tissue transfer (Fig. 2). Postoperative mild palatal swelling (fluid collection) resolved spontaneously. At 6-month follow-up, incisions were well-healed, and feeding/speech were unimpaired (Appendix Fig. 5).

Genetic Analysis

Whole-exome sequencing (WES) was performed on peripheral blood samples from the patient and mother (father's sample was unavailable). Sequencing depth was $100\pm 20\times$ with $>95\%$ coverage of coding regions, aligned to GRCh37/hg19. Analysis focused on craniofacial development-related genes [9]. A heterozygous missense variant was identified in PAX7: c.1033G>C (p.Ala345Pro, NM_001135254.1), present in both the patient and asymptomatic mother.

The variant is absent from gnomAD, ExAC, and 1000 Genomes Project databases (minor allele frequency <0.0001). Amino acid Ala345 is highly conserved across species (human, chimpanzee, pig, mouse, rat, chicken, zebrafish, *Xenopus*; as illustrated in Appendix Fig. 6A). Protein structural prediction (SWISS-MODEL, available at <https://swissmodel.expasy.org/interactive/rfngfe/models/>) suggested potential disruption of the PAX7 OAR domain (a transcription factor activation domain; Appendix Fig. 6B, 3A). Ramachandran plots confirmed stable protein folding, with no major conformational changes (Figs. 3B–E). Per ACMG/AMP [the American College of Medical Genetics and Genomics (ACMG) and Association of Molecular Pathology (AMP)] guidelines [10, 11], the variant is classified as a VUS (Table 1).

Discussion

Craniofacial duplication is an extremely rare anomaly, with partial duplication of the maxilla and upper lip (as in this case) reported in only a handful of studies [1–3]. The patient's phenotype aligns with Type II craniofacial duplication per Chen and Noordhoff's classification (partial duplication of stomatodeal structures) and Hamberis et al.'s revised typology [1, 3], confirming the diagnosis.

Etiological Considerations

The pathogenesis of craniofacial duplication remains unclear. Traditional hypotheses focus on embryogenic disruptions, such as notochordal process cell duplication or abnormal neural crest

cell migration (Appendix Table 1) [3, 5, 8]. With advances in molecular genetics, variants in genes regulating craniofacial development (e.g., SHH, TWIST1, ZIC2, GLI3) have been linked to related anomalies, but no definitive genetic cause for isolated craniofacial duplication has been established [12].

PAX7, a member of the paired box transcription factor family, is critical for embryonic development—particularly neural crest cell regulation, cranial bone formation, facial muscle development, and craniofacial patterning [8, 13]. PAX7 variants have been associated with orofacial clefts, craniofacial dysmorphism, and myopathy [13, 14]. Yankee et al. identified PAX7 as one of 34 top craniofacial-biased genes (highest Gini index in craniofacial tissues), supporting its role in facial development [9].

The PAX7 c.1033G>C (p.Ala345Pro) variant identified here resides in the OAR domain, which mediates transcription factor activity and protein-protein interactions [15]. The affected Ala345 residue is highly conserved across species, suggesting functional importance. However, the variant's pathogenicity remains uncertain: (a) Supporting pathogenicity: Rarity in general populations, conservation of Ala345, predicted OAR domain disruption, and PAX7's craniofacial expression bias. (b) Against pathogenicity: Maternally inherited with incomplete penetrance (mother is asymptomatic), lack of functional validation, and no prior reports in craniofacial duplication.

Incomplete penetrance (as seen in the mother) is common in craniofacial anomalies and may result from modifier genes, epigenetic factors, or environmental triggers [16, 17]. Mosaicism (limited maternal cell line involvement) is another possibility but was not tested due to sample constraints.

Clinical Implications

This case provides preliminary molecular clues linking PAX7 to craniofacial duplication, a phenotype not previously associated with this gene. While the variant is a VUS, its rarity and functional predictions warrant further investigation. For clinical practice: (a) Prenatal screening: PAX7 sequencing may be considered in cases of prenatal maxillary/facial mass suspicious for duplication, though causality must be confirmed. (b) Surgical management: Staged repair (cleft lip first, then duplicated tissue excision) is safe and effective, with three-dimensional CT reconstruction aiding surgical planning. (c) Genetic counseling: Families should be informed of the VUS status, incomplete penetrance risk, and need for long-term follow-up.

Limitations

This study has several limitations: (a) Absence of paternal genetic data, precluding analysis of compound heterozygosity or paternal modifier genes. (b) Lack of functional studies (e.g., in vitro transcription assays, animal models) to validate the variant's impact on PAX7 activity. (c) Single-case design, limiting generalizability. Future research should include functional validation of the PAX7 c.1033G>C variant and sequencing of larger craniofacial duplication cohorts to confirm associations.

Future Perspectives

To advance our understanding of PAX7's role in craniofacial duplication and translate these findings into clinical practice, several key directions warrant further investigation: 1. Functional Validation of the PAX7 Variant: 1) In vitro studies: Conduct transcriptional activity assays (e.g., luciferase reporter assays) to evaluate whether the p.Ala345Pro variant impairs PAX7's ability to regulate downstream target genes involved in craniofacial development (e.g., MSX1, RUNX2). Protein-protein interaction studies (e.g., co-immunoprecipitation) could also clarify if the variant disrupts binding to co-factors required for PAX7-mediated transcription. 2) In vivo models: Generate animal models (e.g., zebrafish, mice) harboring the PAX7 c.1033G>C variant to assess craniofacial phenotypes. Zebrafish are particularly suited for rapid screening of craniofacial anomalies, while conditional PAX7-knockin mice could replicate the human variant's tissue-specific effects. These models may also reveal modifier genes or environmental factors that contribute to incomplete penetrance.

2. Genetic Characterization of Larger Cohorts: 1) Multicenter collaborative studies: Assemble a global cohort of craniofacial duplication patients (including partial and complete phenotypes) to screen for PAX7 variants and other candidate genes (e.g., SHH, TWIST1, ZIC2). This will determine if PAX7 variants are recurrent in the disorder or represent a rare, case-specific finding. 2) Family-based studies: Recruit first-degree relatives of affected individuals (including asymptomatic carriers) to investigate segregation patterns and identify modifier loci via genome-wide association studies (GWAS). Paternal sequencing, unavailable in the current case, should be prioritized to exclude compound heterozygosity or paternal contribution to disease.

3. Expansion of Prenatal Screening and Genetic Counseling: 1) Prenatal diagnostic panels: Incorporate PAX7 into existing craniofacial anomaly screening panels for high-risk pregnancies

(e.g., prenatal ultrasound detecting maxillary masses or facial duplication). This requires validating the variant's pathogenicity in larger cohorts but could enable early diagnosis and prenatal counseling. 2) Informed genetic counseling: Develop tailored counseling guidelines for families with PAX7 VUS, emphasizing incomplete penetrance, variable expressivity, and the need for multidisciplinary care (genetics, craniofacial surgery, pediatrics). Counselors should also address the limitations of current genetic testing and the potential for future functional studies to reclassify variants.

4. Exploration of Molecular Mechanisms Underlying Craniofacial Duplication: 1) Signaling pathway analysis: Investigate how PAX7 variants interact with key craniofacial development pathways (e.g., Hedgehog, Wnt, TGF- β). For example, PAX7 may regulate neural crest cell migration or proliferation, and variant-induced dysfunction could lead to abnormal duplication of facial structures. 2) Epigenetic regulation: Explore DNA methylation or histone modification patterns in PAX7 regulatory regions, as epigenetic changes could explain incomplete penetrance (e.g., differential methylation in symptomatic vs. asymptomatic carriers).

5. Clinical Translation of Surgical Innovations: 1) 3D-printed surgical guides: Standardize the use of 3D-CT reconstruction and patient-specific surgical guides for duplicated tissue excision, as demonstrated in this case. This technology improves surgical precision, reduces operative time, and minimizes damage to adjacent structures (e.g., tooth germs). 2) Long-term outcome studies: Conduct prospective follow-up of craniofacial duplication patients to assess functional outcomes (speech, feeding, facial aesthetics) and identify late complications. This will inform refinements to surgical algorithms and postoperative care.

Conclusion

This study presents a rare case of Type II craniofacial duplication (diprosopus) involving the maxilla and upper lip, successfully managed with staged surgical intervention. Through whole-exome sequencing, we identified a heterozygous missense variant in the PAX7 gene (c.1033G>C, p.Ala345Pro) shared by the patient and his asymptomatic mother. Classified as a VUS per ACMG/AMP guidelines, this variant resides in the functionally critical OAR domain of PAX7— a transcription factor with well-established roles in neural crest cell regulation, craniofacial bone formation, and facial patterning.

Key findings underscore three critical points: First, the patient's phenotype aligns with established craniofacial duplication classifications, confirming the rarity and clinical complexity

of this anomaly. Second, PAX7, previously linked to orofacial clefts and myopathy, is now associated with craniofacial duplication for the first time, expanding the gene's known phenotypic spectrum. Third, the variant's rarity in general populations, high conservation of the affected amino acid, and predicted functional impact collectively support its potential relevance to disease pathogenesis, despite incomplete penetrance in the maternal carrier.

While definitive causality cannot be established in a single case, this report provides preliminary molecular evidence linking PAX7 to craniofacial duplication, addressing a long-standing gap in the genetic understanding of this extremely rare birth defect. The successful surgical management—featuring staged cleft repair, 3D CT-guided duplicated tissue excision, and soft tissue reconstruction—also offers a practical clinical algorithm for similar cases, emphasizing the value of personalized preoperative planning and long-term follow-up.

Supplemental Material 1. This case of craniofacial duplication reported following CARE checklist.

Appendix Videos 1–2. Three-dimensional CT reconstruction of duplicated maxillary arches.

Appendix Table 1. Proposed etiological hypotheses for craniofacial duplication.

Appendix Figs. 1–5. Clinical and radiological images of the patient (preoperative, intraoperative, postoperative).

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Author's contributions

C. Li and D. Huang contributed to study concept and design. C. Li and Z. Gong performed the data collection. C. Li drafted and drew pictures. C. Li edited the English version. All authors contributed to the article and approved the submission.

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Data availability

Data are provided within the manuscript or supplementary information.

Declarations

Ethics approval and consent to participate

This single-patient case report received ethics approval from the Ethics Committee of the First Affiliated Hospital of Xinjiang Medical University (approval no. K20240919-06); approval No. K20240919-06. Written informed consent for publication was obtained from the patient's legal guardian.

Consent for publication

The authors confirm that the patient's legal guardian has signed written informed consent authorizing the use of their personal or clinical details, as well as any de-identifying images for the publication of this study.

Competing interests

The authors declare no competing interests with respect to the research, authorship, and/or publication of this article.

References

1. Hamberis AO, Macias D, Clemmens C, Patel KG. Duplication of the oral cavity and mandible: a rare

- congenital craniofacial anomaly. *BMJ Case Rep.* 2020;13(5):e233799. <https://doi.org/10.1136/bcr-2019-233799>.
2. Costa MA, Borzabadi-Farahani A, Lara-Sanchez PA, Schweitzer D, Jacobson L, Clarke N, et al. Partial craniofacial duplication: a review of the literature and case report. *J Craniomaxillofac Surg.* 2014;42(4):290-6. <https://doi.org/10.1016/j.jcms.2013.05.016>.
 3. Chen YR, Noordhoff MS. Duplication of stomatodeal structures: report of three cases with literature review and suggestion for classification. *Plast Reconstr Surg.* 1989;84(5):733-40.
 4. Sjamsudin J, David D, Singh GD. An Indonesian child with orofacial duplication and neurocristopathy anomalies: case report. *J Craniomaxillofac Surg.* 2001;29(4):195-7. <https://doi.org/10.1054/jcms.2001.0221>.
 5. Cameron AC, McKellar GM, Widmer RP. A case of neurocristopathy that manifests facial clefting and maxillary duplication. *Oral Surg Oral Med Oral Pathol.* 1993;75(3):338-42. [https://doi.org/10.1016/0030-4220\(93\)90147-v](https://doi.org/10.1016/0030-4220(93)90147-v).
 6. Li CX, Huang DS, Gong ZC. Duplicated, Translocated Upper Lip and Maxilla: An Extremely Rare Congenital Craniofacial Anomaly With Novel Genetic Findings. *J Oral Pathol Med.* 2026. <https://doi.org/10.1111/jop.70117>. Epub ahead of print.
 7. Trevisani V, Balestri E, Napoli M, Caraffi SG, Baroni MC, Peluso F, et al. Diprosopus: A Rare Case of Craniofacial Duplication and a Systematic Review of the Literature. *Genes (Basel).* 2023;14(9):1745. <https://doi.org/10.3390/genes14091745>.
 8. Li CX, Gong ZC, Zhang XH, Aelxiding Tuersun, Zhang L. A rare case of complete unilateral cleft lip complicated with maxillary duplication: Apropos of a case report and systematic literature review. *Chin J Stomatol.* 2025;19(2):119-23. <https://doi.org/10.3877/cma.j.issn.1674 - 1366.2025.02.006>.
 9. Yankee TN, Oh S, Winchester EW, Wilderman A, Robinson K, Gordon T, et al. Integrative analysis of transcriptome dynamics during human craniofacial development identifies candidate disease genes. *Nat Commun.* 2023;14(1):4623. <https://doi.org/10.1038/s41467-023-40363-1>.
 10. Wilcox EH, Webb RF, Tshering KC, Hughes MY, Cavé H, DiStefano MT, et al. Updated ACMG/AMP specifications for variant interpretation and gene curations from the ClinGen RASopathy expert panels. *Genet Med Open.* 2025;3:103430. <https://doi.org/10.1016/j.gimo.2025.103430>.
 11. Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genet Med.* 2015;17(5):405-24. <https://doi.org/10.1038/gim.2015.30>.
 12. Haendel M, Vasilevsky N, Unni D, Bologna C, Harris N, Rehm H, et al. How many rare diseases are there? *Nat Rev Drug Discov.* 2020;19(2):77-78. <https://doi.org/10.1038/d41573-019-00180-y>.

13. Leslie EJ, Taub MA, Liu H, Steinberg KM, Koboldt DC, Zhang Q, et al. Identification of functional variants for cleft lip with or without cleft palate in or near PAX7, FGFR2, and NOG by targeted sequencing of GWAS loci. *Am J Hum Genet.* 2015;96(3):397-411. <https://doi.org/10.1016/j.ajhg.2015.01.004>.
14. Quinelato V, Bonato LL, Vieira AR, Granjeiro JM, Menezes K, Borojevic R, et al. PAX7 gene polymorphism in muscular temporomandibular disorders as potentially related to muscle stem cells. *BMC Musculoskelet Disord.* 2021;22(1):959. <https://doi.org/10.1186/s12891-021-04846-w>.
15. Monsoro-Burq AH. PAX transcription factors in neural crest development. *Semin Cell Dev Biol.* 2015;44:87-96. <https://doi.org/10.1016/j.semcdb.2015.09.015>.
16. Ma C, Chen N, Jolly A, Zhao S, Coban-Akdemir Z, Tian W, et al. Functional characteristics of a broad spectrum of TBX6 variants in Mayer-Rokitansky-Küster-Hauser syndrome. *Genet Med.* 2022;24(11):2262-73. <https://doi.org/10.1016/j.gim.2022.08.012>.
17. Edwards SD, Schulze KV, Rosenfeld JA, Westerfield LE, Gerard A, Yuan B, et al. Clinical characterization of individuals with the distal 1q21.1 microdeletion. *Am J Med Genet A.* 2021;185(5):1388-98. <https://doi.org/10.1002/ajmg.a.62104>.

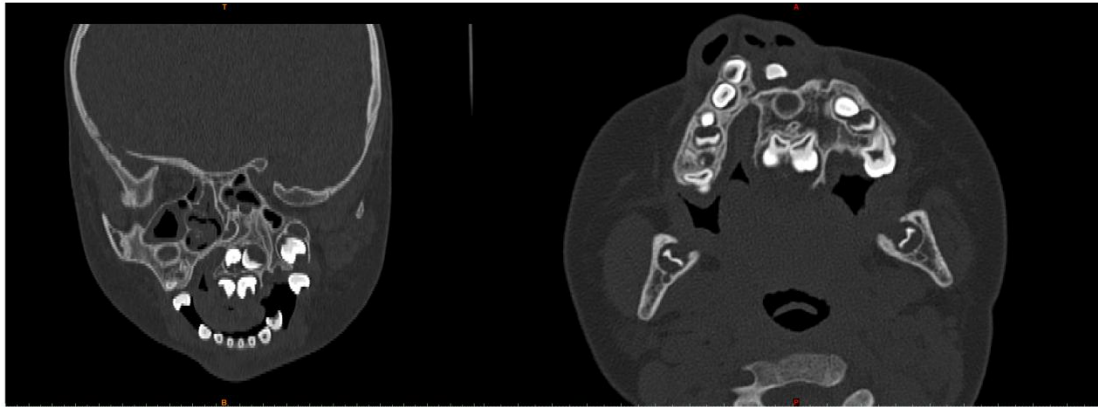
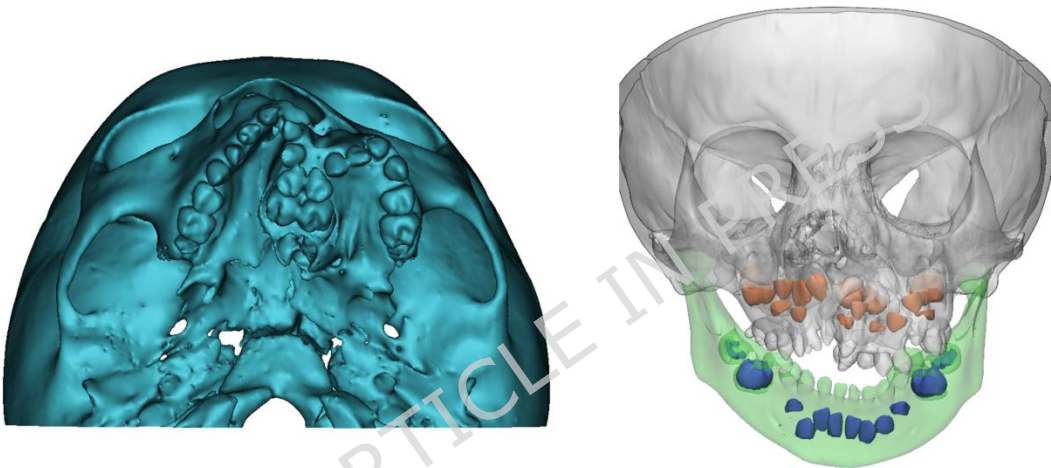
Figure legends**A****B**

Figure 1. Preoperative imaging data for stage II surgery.

Panel A displays coronal (left) and transverse (right) views, showing complete duplication of the maxilla during tooth eruption.

Panel B presents a three-dimensional parametric model generated using Mimics software version 19.0 (Materialise Inc., Leuven, Belgium) for three-dimensional plane reorientation and reconstruction.

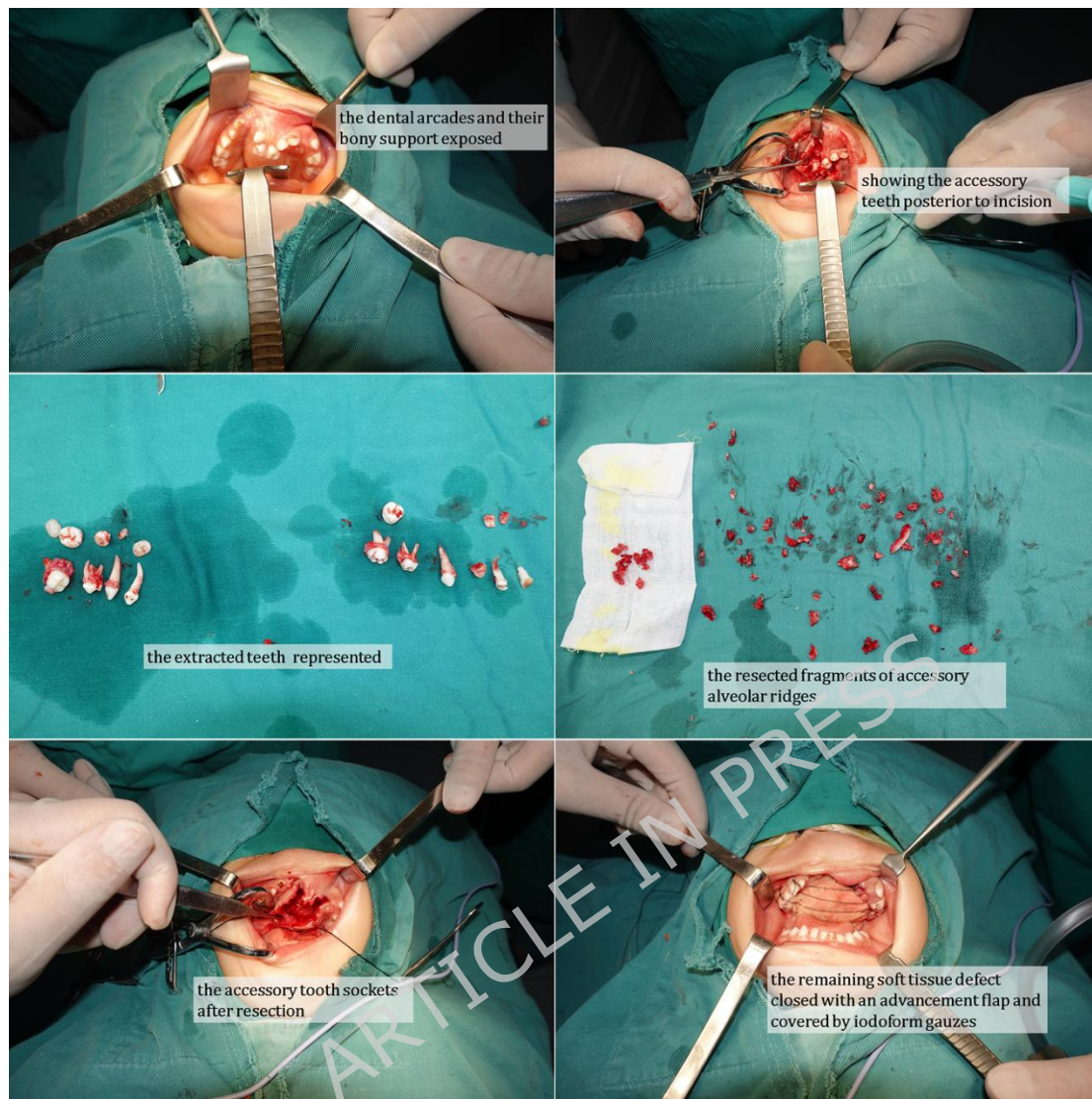


Figure 2. Intraoperative photographs illustrating the duplicated maxilla before incision, during the resection procedure, and the restored accessory tooth sockets following resection.

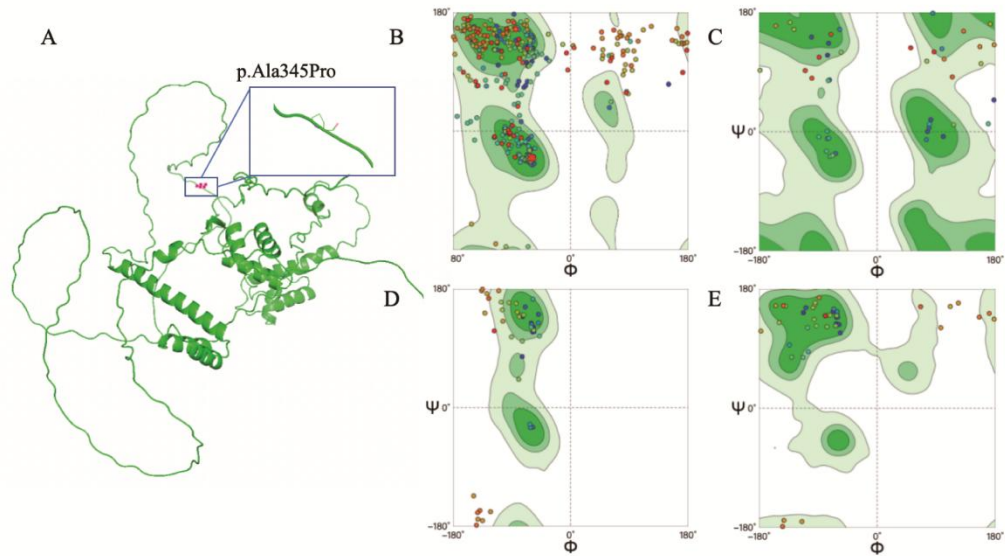


Figure 3. PAX7 protein structural analysis. A. SWISS-MODEL prediction of PAX7 (wild-type vs. p.Ala345Pro variant). The box indicates OAR domain structural alterations in the variant. B-E. Ramachandran plots showing ϕ/ψ dihedral angles for all residues (B), glycine (C), proline (D), and pre-proline (E), confirming stable protein folding.

Table 1. ACMG/AMP Classification of PAX7 c.1033G>C (p.Ala345Pro) Variant

ACMG/AMP Criterion Category	Specific Criterion	Finding	Classification Impact
Population Allele Frequency	Absence from general population databases	Not present in gnomAD, ExAC, or 1000 Genomes (MAF < 0.0001)	Supporting pathogenicity
Sequence Conservation	Cross-species conservation of the affected amino acid	Ala345 is fully conserved across 8 species (human, chimpanzee, pig, mouse, rat, chicken, zebrafish, Xenopus)	Supporting pathogenicity
Functional Predictions	Impact on protein structure/function	Predicted to disrupt the OAR domain (critical for transcription factor activity) via SWISS-MODEL analysis	Supporting pathogenicity
Inheritance Pattern	Segregation with disease in the family	Maternally inherited; mother is asymptomatic (incomplete penetrance)	Against pathogenicity
Clinical Phenotype Correlation	Prior reports of variant in relevant phenotypes	No previous associations with craniofacial duplication or related anomalies	Against pathogenicity
Functional Studies	Experimental validation of variant impact	Not performed (no in vitro/in vivo functional assays)	Neutral
Final Classification	Synthesis of criteria per ACMG/AMP 2015 guidelines	Variant of Uncertain Significance (VUS)	—

Notes:

1. Classification follows the 2015 ACMG/AMP guidelines for interpreting sequence variants [10].
2. "MAF" = Minor Allele Frequency; "OAR domain" = Octapeptide/Alanine-Rich domain (critical for PAX7 transcriptional activation).
3. Incomplete penetrance (asymptomatic carrier mother) is a common phenomenon in craniofacial developmental anomalies, contributing to the VUS classification.